Nisadu Nimsitha – CV

Cloud & DevOps Engineer | Backend & Distributed Systems | Computer Science Undergraduate

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Professional Summary

Computer Science undergraduate at the University of Sri Jayewardenepura (GPA 3.31/4.00) focused on cloud infrastructure, DevOps automation and distributed backend systems. Deployed six Spring Boot microservices on AWS EC2 with k3s, Amazon RDS PostgreSQL and AWS CodePipeline CI/CD, load-tested with k6 at 100 concurrent users and 0% failures. Hands-on with Terraform, GitHub Actions (OIDC, Trivy), Prometheus and Grafana.

Technical Skills

- **Cloud & Infrastructure:** AWS (EC2, VPC, ECR, RDS PostgreSQL, CodePipeline, CodeBuild, IAM, S3, KMS), Kubernetes (k3s), Docker, Docker Compose, Linux (Ubuntu, Bash, systemd, SSH)
- **DevOps & CI/CD:** Terraform, GitHub Actions (OIDC, Trivy), AWS CodePipeline, AWS CodeBuild, Jenkins, Maven, Git
- **Backend:** Java, Spring Boot (Spring Security, Spring Data JPA), Python, FastAPI, Node.js, Express, REST APIs, Microservices, JWT Authentication
- **Observability & Testing:** Prometheus, Grafana, k6, pytest, Postman
- **Databases:** PostgreSQL, MySQL, Database Design & ER Modeling
- **Machine Learning & MLOps:** Scikit-learn, XGBoost, MLflow
- **Frontend:** React, Next.js, Vite, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS
- **Networking & Security:** TCP/IP, Subnetting (VLSM), OSI Model, Cisco Packet Tracer, VPC Security Groups & NACLs

Education

University of Sri Jayewardenepura, Colombo, Sri Lanka | 2024 – Expected 2026

Bachelor of Science (Honours) in Computer Science (Statistics & Mathematics) | GPA: 3.31 / 4.00

- Coursework: Service-Oriented Computing, Data Structures & Algorithms, Computer Security, Machine Learning, Database Design & ER Modeling, Net-Centric Computing, Software Quality Assurance, Object-Oriented Programming, Regression Analysis, Inference Statistics

Projects

Lanka MicroJob – Distributed Cloud Microservices Platform

Spring Boot, React, AWS EC2, k3s, Amazon ECR, AWS CodePipeline, Amazon RDS, k6 | GitHub | Architecture case study | Live demo

- Engineered six Spring Boot microservices (API Gateway, User, Job, Matching, Broker, Notification) covering authentication, job placement, commission logic and audit trails
- Containerized the services with Docker and deployed them to a single-node k3s cluster on AWS EC2 using YAML manifests, ClusterIP services, NodePort ingress and an Elastic IP
- Built an AWS CodePipeline and CodeBuild workflow, triggered by GitHub webhooks, that runs Maven tests, builds Docker images, pushes to Amazon ECR and performs rolling updates
- Integrated Amazon RDS PostgreSQL in an isolated VPC subnet with connection pooling; k6 tests sustained 100 peak virtual users with 0% failures and about 480 ms P95 latency
- Resolved CORS preflight issues and JWT propagation through the API Gateway; added health endpoints for Kubernetes liveness and readiness probes

CloudOps – Automated Cloud Deployment & Reliability Platform

Terraform, AWS (VPC, IAM, EC2, ECR), Docker, GitHub Actions, Prometheus, Grafana, FastAPI, Python

- Wrote modular Terraform to provision reproducible AWS infrastructure: custom VPCs, public and private subnets, internet gateways, route tables, security groups and Amazon ECR repositories
- Designed a GitHub Actions pipeline using AWS OIDC federation to remove long-lived IAM credentials, with Trivy scans that gate releases on container vulnerabilities
- Instrumented a FastAPI service with Prometheus RED metrics (rate, errors, duration) and latency histograms, and built Grafana dashboards for latency and error tracking
- Added pytest suites to the pre-merge CI workflow and wrote rollback runbooks for deployment regressions

Dengue Outbreak Risk Classifier & MLOps Pipeline

Python, Scikit-learn, XGBoost, MLflow, FastAPI, GitHub Actions | GitHub

- Built an end-to-end outbreak risk classifier on historical case counts and weather data, splitting 13,150 training and 2,800 validation records chronologically to prevent target leakage
- Engineered rolling lag features and weather moving averages; compared Logistic Regression, Random Forest and XGBoost, reaching a validation macro F1 of 0.7888 and 0.8437 recall on the high-risk class
- Tracked experiments with MLflow and served the champion model through an asynchronous FastAPI endpoint with automated GitHub Actions tests

Lanka Travel Planner – AI-Powered Travel Engine

React, Node.js, Express, Google Gemini API, Leaflet, OpenStreetMap | GitHub

- Built a full-stack planner that generates multi-day Sri Lankan itineraries from destination preferences, budget and traveler profile, with Gemini-generated cost breakdowns and daily schedules
- Rendered interactive route maps with Leaflet and OpenStreetMap, including waypoints and location popups

Experience & Leadership

Web Developer & Lead Graphic Designer, CODEXIA 2026

Computer Science Association (CSA), University of Sri Jayewardenepura | 2026 | Event platform

- Co-developed the event platform with Next.js, React and Tailwind CSS, including a canvas starfield and mobile-responsive layouts
- Produced the official media kit: promotional campaigns, formal invitations and participant and winner certificates
- Worked with the CSA committee on competition logistics, problem guidelines and technical infrastructure

Web Development Team Member, Sathsara Dahana 2026

Faculty of Applied Sciences, University of Sri Jayewardenepura | 2026 | Event platform

- Built the official portal for the faculty's annual cultural festival with Next.js, React and Tailwind CSS
- Implemented an audition verification system that lets candidates check selection results by index number, plus an HTML5 canvas terminal bootloader and live countdowns

Customer Care Executive, Startek

Colombo, Sri Lanka

- Resolved tier-1 technical and customer issues for global enterprise accounts under strict SLAs, using structured root-cause troubleshooting

Certifications

- **AWS Academy Graduate:** Cloud Foundations (May 2026), Cloud Security Foundations (Jul 2026), Microservices & CI/CD Pipeline Builder (Aug 2026), Cloud Security Builder (Sep 2026)
- **Cisco Networking Academy:** Networking Basics (Oct 2026), Introduction to Packet Tracer (Sep 2026)
- **KodeKloud:** Kubernetes Crash Course (May 2026), Linux for Absolute Beginners (May 2026)
- **In progress:** Linux Essentials (Cisco / LPI), GitOps for Platform Engineering (KodeKloud)